

— A DOCUMENT UNDERSTANDING PROJECT · ACCOUNTS PAYABLE

Invoice Processing.

One scenario, end to end · Read the page · Match three documents · Decide

• 12 INVOICES • 5 LEVELS • 15 RULES • ONE DECISION EACH

– PROJECT / CONTENTS

What we'll cover.

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- 01 **The process** — what accounts payable does all day.
 - 02 **The document** — an invoice is a picture, not a form.
 - 03 **The pipeline** — read, match, decide, and where each breaks.
 - 04 **The fields** — what you extract, and what is checked.
 - 05 **The match** — three documents that have to agree.
 - 06 **Build it live** — then your own bot, scored.
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– 01 / EXPECTED TO LEARN

By the end of today, you can —

- 01 Scope a DU project: **the document, the fields, the decision.**
- 02 Read an invoice at **five levels of damage** — and say where yours stops.
- 03 Run a three-way match and read a violation **by its rule ID.**
- 04 Approve, reject or hold — and **defend which one.**

– 02 / THE PROCESS

– AS-IS · A PERSON

Ninety seconds an invoice

Opens the PDF and reads it

Finds the order and the receipt

Compares price, quantity and tax

Keys the result into the system

Repeats until the queue is empty

– TO-BE · A BOT

The same ninety seconds, once

Claims one item from the queue

Extracts the header and the lines

Submits, then reads the match result

Approves, rejects or holds

Closes the item with its reason

- THE HARD PART WAS NEVER THE CLICKING

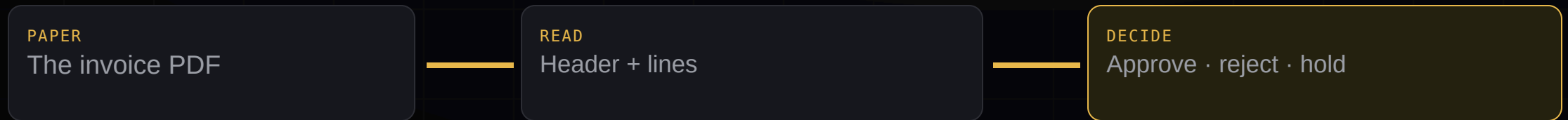
– 02.1 / WHY IT IS HARD

Three problems. Only one is reading.

- A tutorial invoice puts every field where you expect it. A real one is a photograph.
- A third of these invoices are wrong, and the wrong ones are the ones that matter.
- The truth is not on the page. It is on the page, the order and the receipt.

– 03 / THE PIPELINE

Read it. Match it. Decide it.



- Each stage fails differently — a misread digit is not a wrong decision.
- They are scored separately, so you can tell which stage cost you.

Build all three or you have built none: a reader that approves a duplicate has failed.

– 04 / THE DOCUMENT

Three regions

Every supplier invoice in the lab has the same three parts and prints them differently: about half the vendors print bilingual, a third print Arabic first, and two in five write their numbers in Eastern Arabic digits. Your reader has to find the parts, not the pixels.

hdr The header — who billed, against which order, in which currency. Twelve fields.

lin The lines — code, quantity, unit, price, tax, total. Eight fields on each one.

tot The totals — printed on the page, and not always equal to the sum of the lines.

— 04.1 / WHAT YOU EXTRACT

Twelve header fields, eight per line.

- | | |
|----|--|
| 01 | Identity — number, date, due date, and the vendor as printed on the page. |
| 02 | Reference — the purchase order number. Without it there is nothing to match. |
| 03 | Money — currency, subtotal, tax, grand total. Read them; do not compute them. |
| 04 | Payment — IBAN and bank name. The most expensive field on the page. |

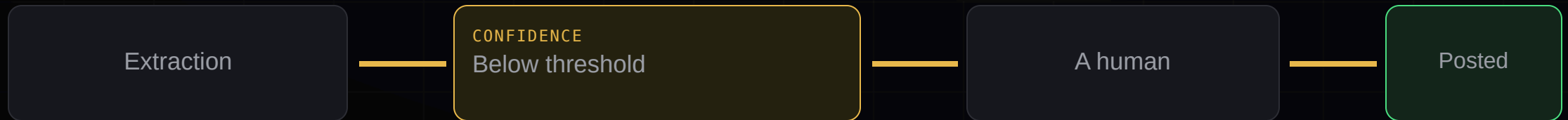
– 04.2 / THE LADDER

The same invoice, five times harder.

- | | |
|----|---|
| 01 | Native PDF — vector text, selectable. No OCR needed. |
| 02 | Clean scan — 300 dpi, faint blur, a fraction of a degree of skew. |
| 03 | Office scan — 200 dpi, three degrees of skew, noise, uneven light. |
| 04 | Phone photo — perspective, a shadow gradient, a warm cast, soft focus. |
| 05 | Handled document — stamps, handwriting, staple marks, fold lines. |

– 05 / VALIDATE

A confidence score is a routing decision.



- High confidence posts straight through; low confidence goes to a person.
- Threshold too high and a human does the work. Too low and the wrong number is paid.

The lab gives you no confidence to hide behind: you submit what you read, and it is right or not.

– 06 / THE THREE-WAY MATCH

Three documents. One version of events.

ORDER

What we agreed

price · quantity · vendor

RECEIPT

What arrived

quantity · date · condition

INVOICE

What they billed

price · quantity · tax

Every violation carries a rule ID – PO-INV-PRICE, GRN-QTY, BANK-CHANGE. Branch on the ID, never on the message text: the message is written for a person and may change.

– 07 / THE DECISION

Approve, reject, or refuse to pay.

- ✓ **Clean** — approve it and pay it. A warning is approved with a note, not held.
- ✗ **Error** — reject it. Wrong price, wrong quantity, no purchase order: it goes back.
- ✗ **Critical** — never pay it. A duplicate, or a changed bank account, is an exception.

Holding a clean invoice costs you as much as paying a bad one. The rubric counts both.

■ LIVE · TOGETHER

What we'll do right now.

- ① Open the queue and claim one invoice — twelve are waiting.
- ② Read it at level 1, submit the header and lines, watch the match run.
- ③ Open the same invoice at level 4 and read it again. Watch what your eyes do.
- ④ Break it on purpose: change the IBAN by one digit, and watch BANK-CHANGE fire.

– 08 / YOUR TASK

Automate the twelve.

DO

Practise until the match result holds no surprises at level 1.

Move to level 3 and practise again — the reading is the hard part.

Take one scored run. Finish it, whatever the number says.

Note the rule ID that cost you the most items.

DELIVER – BEFORE THE NEXT SESSION

Your certificate code, or your score if you did not pass.

The header fields your reader got wrong, and at which level.

One sentence: where would you set the confidence threshold, and why.

– 08.1 / INTERVIEW SEED

If they ask: “Walk me through an invoice automation.”

- 1 **Name the documents** — invoice, purchase order, goods receipt. Three-way match.
- 2 **Name the reading** — five levels of scan quality, bilingual, Eastern Arabic digits.
- 3 **Name the control** — every violation has a rule ID, and the bot branches on the ID.
- 4 **Name the exception** — what your bot refuses to decide, and who it hands it to.

- THE EXCEPTION PATH IS THE ANSWER THEY ARE LISTENING FOR

– 09 / RECAP

A document, a match, a decision.

- ✓ An invoice is three regions and twenty-odd fields, printed differently by every vendor.
- ✓ A three-way match turns three documents into one answer, with a rule ID behind it.
- ✓ Approve, reject or hold — and the hold is a decision you have to defend.

– 09.1 / WHAT'S MISSING

One scenario is not a pipeline.

Invoice processing is the deepest of the four, but it is still one queue and one document type. Classification, routing between types, and a production exception queue are the next session. Bring the rule ID that cost you the most.

– 10 / NEXT

Bring me twelve invoices.

NEXT → Your scored run on invoice-processing

the breakdown · the rule that fired most · where you would set the threshold

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